Course Syllabus

Hazard Planning and Risk Management

Spring 2026

Course Code: Not Defined

Course Day/Time: Wed and Thurs (10:00 am-11:30 am)

Prerequisites: None

Class format: Two classes per week of 1.5 hours each

Room: GIS Lab USPCASW

Instructor: Arjumand Zaidi, Ph.D.

The instructor has professional experience in the field of environmental evaluation and decision-making. Her research interests include optimizing and modeling water resources and environmental and disaster management systems. Most of her research work deals with environmental decision-making with the help of various numerical techniques and Geographical Information Systems (GIS) using satellite data.

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Course Goal

To enable students to critically understand hazard planning and disaster management processes, as well as their challenges and achievements in the context of the Pakistan disaster management system.

Course Summary:

In recent years, disasters caused by natural hazards have affected human societies more than ever. These phenomena are intensified not only due to climatic changes, loss of biodiversity, exploitation of natural resources, and human-induced activities. The vulnerability of humankind has also increased, making these natural phenomena become disasters. The course explores the relationship that makes people more vulnerable and increases the frequency of natural hazards and corresponding disasters.

This course will give students comprehensive hazard planning and management understanding and skills. The course will cover topics related to *Hazard Mapping, Risk and Vulnerability Analysis, and Evacuation Planning* using GIS and spatial analytical software to address different hazard and disaster management phases. Detailed course contents are presented in the "Course Contents" table.

Objectives:

The primary objectives of this course are:

1. To become familiar with the basic concepts of natural hazards and related disasters.
2. To demonstrate how human landuse activities create hazards and, in this respect, examine the physical environment and human behavior.
3. To introduce planning tools, including geospatial techniques that can be used to mitigate the hazards, including education programs for the vulnerable community, public awareness, identification of hazard zones and land use planning within these zones, structural control measures, evacuation procedures, and planning techniques, and introduction of crowdsourcing and use of internet-based information dissemination.
4. To develop skills for designing a disaster management project.

Learning outcomes:

On completion of this course, students will be able to;

1. **LEARN** the concepts of hazards and disasters (H&D) management.
2. **UNDERSTAND** the causes and characteristics of natural hazards and related disasters.
3. **DISTINGUISH**– between different phases of H&D management and
4. **APPLY** hazard management knowledge to recognize, assess, and deal with disaster risks.
5. **LEARN** meaningful disaster prevention, mitigation, response, and recovery strategies.
6. **USE** GIS and satellite data in different phases of hazard management.
7. **DEVELOP** project on hazard and disaster management using geospatial techniques.

8. **Course Contents**

SN	Contents
Week 1 & 2	**Introduction:** Introduction to the class and the course (purpose, application in the real world, etc.) **Disaster Principles, Introduction, Definition, Basic Concepts:** Hazard classification, difference between disaster, emergency, and hazard; world disasters; types of hazards and disasters; disaster risk; disaster risk management, disaster characteristics; effects of disasters; disasters in developing/developed countries; Pakistani scenario.
Week 3	**Role of Remote Sensing and GIS for Disaster Management:** Post Disaster and Pre-Disaster: Evaluate spatial data requirements in disaster risk management (e.g., DEMs, topo data, population data, flood data, earthquake data, etc.)
Week 4, 5 & 6	**Disaster Management Cycle:** Mitigation, Response, Recovery, and Rehabilitation – definitions, tools (Legislation, Early Warning Systems, Emergency Planning, Disaster management plan, Insurance, Public Awareness, Training, and Education), reporting, etc. **Roles and responsibilities for disaster planning and response :**
Week 7	**Risk and Vulnerability Analysis and Assessment:** definitions, purpose, risk analysis steps, hazard identification, generic risk assessment model, qualitative and quantitative risk analyses, risk matrix, risk reduction, risk curves, elements at risk, sources of element at risk matrix
Week 8	**Geological Disasters:** Tectonic Hazards I: Earthquake (causes, characteristics, prediction, and mitigation) **Midterm Exam**
Week 9 & 10	**Overview of the Midterm exam** **Mid-semester project review** **Geological Disasters:** Tectonic Hazards II: Tsunami (generation, movement, observation, and mitigation) **GIS Hands-on**: Tsunami mapping using DEM and GIS (exercise)
Week 11	**Hydrologic Disasters I** 1. Severe Storms and hurricanes (causes, damages, monitoring, and planning) 2. Droughts: Meteorological, Agricultural, Hydrological, Socioeconomic (famine). Causes, monitoring, mitigation, and management
Week 12	**Hydrologic disaster II** 1. Floods (types, processes, simulation and prediction, frequency analysis) **Rapid Flood mapping using satellite data and GIS** (Exercise)
Week 13	**National Disaster Risk Management Framework - Pakistan** **Roles and responsibilities for disaster planning and response – II:** Role of government at district, provincial, and national levels – Guest lecture by PDMA (provincial disaster management authority) or NDMA representative

Week 14	**Damage/loss Assessment:** Need, cost of a disaster, direct and indirect losses, Guidelines Established by FEMA, a brief intro of HAZUS, population estimates
Week 15	**Project/Poster Presentations**
Week 16	**Final Exam**

Reference Books:

1. Environmental hazards: assessing risk and reducing disaster By Keith Smith, David N. Petley.
2. Introduction to International Disaster Management By Damon P. Coppola, Third Edition 2015 (access through http://www.digitallibrary.edu.pk/)
3. Introduction to Emergency Management By George D. Haddow, Bullock, and Coppola (Soft copy of fourth edition available)
4. Successful Response Starts with a Map: Improving Geospatial Support for Disaster Management - Committee on Planning for Catastrophe: A Blueprint for Improving Geospatial Data, Tools, and Infrastructure, National Research Council - (PDF version – Will be provided)
5. Remote Sensing for Hazard Monitoring and Disaster Assessment: Marine and Coastal Applications in the Mediterranean Region (Current Topics in Remote Sensing) By Eric Charles, Krystyna A. Brown, Anton Micalle

Websites:

The instructor should encourage students to become familiar with the following sites;

1. http://www.unisdr.org/: United Nations International Strategy for Disaster Reduction
2. www.ifrc.org/: World Disasters Report
3. https://wg1.ipcc.ch/srex/: Report on Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation
4. www.fema.gov: Information on the federal emergency management system of the USA.
5. www.colorado.edu/hazards: sharing knowledge concerning the social science and policy aspects of disasters occurring in the US and internationally.

Field Trips

1. Possibly, a field trip to PMD Karachi will be conducted by the middle of the semester.

Disaster Reports (Assignments)

1. Every student will participate in reporting/updating the disasters and their details occurring worldwide during this semester. The detail may include information on the type, effects, death/injury toll, and economic impact of each disaster.

Project:

1. To design and implement an RS/GIS-based project integrating remote sensing data, GIS tools, and Hazards Vulnerability and Risk (HVR) models. Experts will evaluate project presentations.
2. Write a short paper based on your project (follow the IEEE-IGARSS conference format).
3. We will submit selected papers to any upcoming conference.

Distribution of Marks:

Quizzes	10 %
Midterm	25 %
Assignments	10 %
Field Trips	
Participation	2.5 %
Field Trip Report	2.5 %
Final Exam	50 % (including project marks)

Attendance Policy: As per USPCAS-W guidelines/policy, 90% attendance is required except in some genuine, verifiable emergencies. If you have a specific problem with attendance, notify the instructor before class unless the emergency is such that this is impossible. Consistent with University guidelines, students absent from regularly scheduled examinations because of authorized University activities will have the opportunity to take them at an alternate time. Make-up exams for absences due to other reasons will be at the instructor's discretion.

Policy on Academic Honesty: All work submitted by students should be their own. Plagiarism of any type will not be tolerated and will result in F- grade. Students are highly recommended to attend a workshop on plagiarism, its types, and ways to avoid plagiarism (to be scheduled soon).

Student Honor Code for Exams: Students must sign the honor code document before every exam (provided by the instructor).

Schedule of Assignments: Assignments will be given to students throughout the semester with well-defined due dates.

Policy on missed or late homework: Homework should be submitted on its due date given with each assignment. Each late day will cost two (02) point deductions on the total 10 points assigned to each HW till the total points become zero.

Policy on Academic Misconduct/Dishonesty: There is a zero-tolerance policy for any form of academic dishonesty. Students found engaging in plagiarism, cheating, or forgery during any assignment or test will be subject to the conduct code policies of the University.

Social Justice Statement: "Mehran University of Engineering & Technology (MUET)" is committed to social justice. I concur with that commitment and expect to foster a nurturing learning environment based on open communication, mutual respect, and non-discrimination. Our University does not discriminate based on race, sex, age, disability, veteran status, religion, sexual orientation, color, or national origin. Any suggestions on how to further such a positive and open environment in this class will be appreciated and seriously considered.

Disability accommodations: The University provides an equal educational opportunity to students with disabilities otherwise qualified and equal access to University programs and activities provided to other students.